多元复合函数的求导法则

复习：一元复合函数的求导法则

$y=f(u)$，$u=\varphi(x)$ $\Rightarrow$ $y=f[\varphi(x)]$

$$\frac{dy}{dx}=\frac{dy}{du}\cdot\frac{du}{dx}=f'(u)\cdot\varphi'(x)=f'[\varphi(x)]\cdot\varphi'(x)$$

$$\begin{tikzcd} y \arrow[r, "\frac{dy}{du}"'] & u \arrow[l, bend left, "\frac{dy}{du}\cdot\frac{du}{dx}"] \arrow[r, "\frac{du}{dx}"'] & x \end{tikzcd}$$

一、中间变量为一元的情形

$z=f(u,v)$，$u=\varphi(t)$，$v=\psi(t)$ $\Rightarrow$ $z=f[\varphi(t),\psi(t)]\triangleq z(t)$

设 $z=f(u,v)$，$u=\varphi(t)$，$v=\psi(t)$，

其中 $\varphi(t)$，$\psi(t)$ 在 t 处可导，$f(u,v)$ 在对应点 (u,v) 处可微，

则复合函数 $f[\varphi(t),\psi(t)]$ 在 t 处可导。且：

$$\begin{tikzcd} & z & \\ u \arrow[ur] & & v \arrow[ul] \\ & t \arrow[ul] \arrow[ur] & \end{tikzcd}$$

$$\frac{dz}{dt}=\frac{\partial f}{\partial u}\cdot\frac{du}{dt}+\frac{\partial f}{\partial v}\cdot\frac{dv}{dt}$$

$$=\frac{\partial z}{\partial u}\cdot\frac{du}{dt}+\frac{\partial z}{\partial v}\cdot\frac{dv}{dt}$$

proof:

（目标：求 $\lim\limits_{\Delta t\to 0}\frac{\Delta z}{\Delta t}$）

$\because$ $f(u,v)$ 在点 (u,v) 处可微

$\therefore$ $\Delta z=\frac{\partial f}{\partial u}\cdot\Delta u+\frac{\partial f}{\partial v}\cdot\Delta v+o\left(\sqrt{(\Delta u)^2+(\Delta v)^2}\right)$

同除以 Δt，

$$\frac{\Delta z}{\Delta t}=\frac{\partial f}{\partial u}\cdot\frac{\Delta u}{\Delta t}+\frac{\partial f}{\partial v}\cdot\frac{\Delta v}{\Delta t}+\frac{o\left(\sqrt{(\Delta u)^2+(\Delta v)^2}\right)}{\Delta t}$$

$$=\frac{\partial f}{\partial u}\cdot\frac{\Delta u}{\Delta t}+\frac{\partial f}{\partial v}\cdot\frac{\Delta v}{\Delta t}+\frac{o\left(\sqrt{(\Delta u)^2+(\Delta v)^2}\right)}{\sqrt{(\Delta u)^2+(\Delta v)^2}}\cdot\frac{\sqrt{(\Delta u)^2+(\Delta v)^2}}{\Delta t}$$

$$=\frac{\partial f}{\partial u}\cdot\frac{\Delta u}{\Delta t}+\frac{\partial f}{\partial v}\cdot\frac{\Delta v}{\Delta t}+\frac{o(\rho)}{\rho}\cdot\underbrace{\sqrt{\left(\frac{\Delta u}{\Delta t}\right)^2+\left(\frac{\Delta v}{\Delta t}\right)^2}}_{(\varphi'(t))^2+(\psi'(t))^2\ \to\ \text{常量}}\cdot\operatorname{sgn}(\Delta t)$$

$\because$ $u=\varphi(t)$，$v=\psi(t)$ 在 t 处可导

$\therefore$ $u=\varphi(t)$，$v=\psi(t)$ 在 t 处连续

$\therefore$ 在 t 处（即点 (u,v) 处），$\Delta u\to 0$ $(\Delta t\to 0)$，$\Delta v\to 0$ $(\Delta t\to 0)$

$\therefore$ $\rho=\sqrt{(\Delta u)^2+(\Delta v)^2}\to 0$ $(\Delta t\to 0)$

$\therefore$ $\lim\limits_{\Delta t\to 0}\frac{\Delta z}{\Delta t}=\frac{\partial f}{\partial u}\cdot\frac{du}{dt}+\frac{\partial f}{\partial v}\cdot\frac{dv}{dt}$

推广：

$$z=f(u,v,w),\ u=\varphi(t),\ v=\psi(t),\ w=\omega(t) \Rightarrow z=f[\varphi(t),\psi(t),\omega(t)] \triangleq z(t)$$

$$\frac{dz}{dt}=\frac{\partial f}{\partial u}\cdot\frac{du}{dt}+\frac{\partial f}{\partial v}\cdot\frac{dv}{dt}+\frac{\partial f}{\partial w}\cdot\frac{dw}{dt}$$

二、中间变量为多元的情形

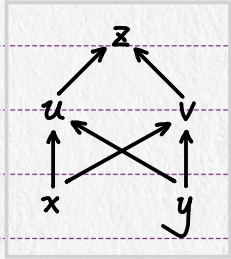
$$z=f(u,v),\ u=\varphi(x,y),\ v=\psi(x,y) \Rightarrow z=f[\varphi(x,y),\psi(x,y)] \triangleq z(x,y)$$

设 $z=f(u,v)$，$u=\varphi(x,y)$，$v=\psi(x,y)$，

其中 φ,ψ 在 (x,y) 处可偏导，$f(u,v)$ 在对应点 (u,v) 处可微，

则复合函数 $z=f[\varphi(x,y),\psi(x,y)]$ 在 (x,y) 处可偏导。且：

$$\begin{cases}\dfrac{\partial z}{\partial x}=\dfrac{\partial f}{\partial u}\cdot\dfrac{\partial u}{\partial x}+\dfrac{\partial f}{\partial v}\cdot\dfrac{\partial v}{\partial x}\\[2ex]\dfrac{\partial z}{\partial y}=\dfrac{\partial f}{\partial u}\cdot\dfrac{\partial u}{\partial y}+\dfrac{\partial f}{\partial v}\cdot\dfrac{\partial v}{\partial y}\end{cases}$$



推广：

$$z=f(u,v,w),\ u=\varphi(x,y),\ v=\psi(x,y),\ w=\omega(x,y) \Rightarrow z=f[\varphi(x,y),\psi(x,y),\omega(x,y)] \triangleq z(x,y)$$

$$\begin{cases}\dfrac{\partial z}{\partial x}=\dfrac{\partial f}{\partial u}\cdot\dfrac{\partial u}{\partial x}+\dfrac{\partial f}{\partial v}\cdot\dfrac{\partial v}{\partial x}+\dfrac{\partial f}{\partial w}\cdot\dfrac{\partial w}{\partial x}\\[2ex]\dfrac{\partial z}{\partial y}=\dfrac{\partial f}{\partial u}\cdot\dfrac{\partial u}{\partial y}+\dfrac{\partial f}{\partial v}\cdot\dfrac{\partial v}{\partial y}+\dfrac{\partial f}{\partial w}\cdot\dfrac{\partial w}{\partial y}\end{cases}$$

推广：

$$g=f(u,v),\ u=\varphi(x,y,z),\ v=\psi(x,y,z) \Rightarrow g=f[\varphi(x,y,z),\psi(x,y,z)] \triangleq g(x,y,z)$$

$$\begin{cases}\dfrac{\partial g}{\partial x}=\dfrac{\partial f}{\partial u}\cdot\dfrac{\partial u}{\partial x}+\dfrac{\partial f}{\partial v}\cdot\dfrac{\partial v}{\partial x}\\[2ex]\dfrac{\partial g}{\partial y}=\dfrac{\partial f}{\partial u}\cdot\dfrac{\partial u}{\partial y}+\dfrac{\partial f}{\partial v}\cdot\dfrac{\partial v}{\partial y}\\[2ex]\dfrac{\partial g}{\partial z}=\dfrac{\partial f}{\partial u}\cdot\dfrac{\partial u}{\partial z}+\dfrac{\partial f}{\partial v}\cdot\dfrac{\partial v}{\partial z}\end{cases}$$

例：设 $z=e^u\sin v$，$u=xy$，$v=x+y$，求 $\frac{\partial z}{\partial x}$，$\frac{\partial z}{\partial y}$。

解：

$$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}$$
$$=e^u\sin v\cdot y+e^u\cos v\cdot 1$$
$$=e^{xy}\cdot\left[y\sin(x+y)+\cos(x+y)\right]$$

$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}$$
$$=e^{xy}\cdot\left[x\sin(x+y)+\cos(x+y)\right]$$

三、抽象复合函数的偏导数

引入记号：

$f(u,v)$
- $f'_u \triangleq f'_1$，$f'_v \triangleq f'_2$
- $f''_{uu} \triangleq f''_{11}$，$f''_{uv} \triangleq f''_{12}$，…
- …

$f(u,v,w)$
- $f'_u \triangleq f'_1$，$f'_v \triangleq f'_2$，$f'_w \triangleq f'_3$
- $f''_{uv} \triangleq f''_{12}$，$f''_{vw} \triangleq f''_{23}$，…
- …

注：若 $f(u,v)$，则 f'_1、f'_2 仍是 u,v 的函数，即：$f'_1(u,v)$、$f'_2(u,v)$。

如：

$$z=f(xy,\sin(x+y))$$

则：

$$f'_1(xy,\sin(x+y))\ 、\ f'_2(xy,\sin(x+y))$$

例：$w=f(x+y+z,xyz)$，求 $\frac{\partial w}{\partial x}$，$\frac{\partial^2 w}{\partial x\partial z}$。

解：

$$\frac{\partial w}{\partial x} = f_1' \cdot 1 + f_2' \cdot yz$$
$$= \underbrace{f_1'}_{f_1'(x+y+z,\ xyz)} + \underbrace{f_2'}_{f_2'(x+y+z,\ xyz)} \cdot yz$$

$$\frac{\partial^2 w}{\partial x \partial z} = \frac{\partial f_1'}{\partial z} + y \cdot f_2' + yz \cdot \frac{\partial f_2'}{\partial z}$$
$$= \left(f_{11}'' \cdot 1 + f_{12}'' \cdot xy \right) + y \cdot f_2' + yz \cdot \left(f_{21}'' \cdot 1 + f_{22}'' \cdot xy \right)$$
$$= f_{11}'' + xy \cdot \underline{f_{12}''} + y \cdot f_2' + yz \cdot \underline{f_{21}''} + xy^2 z \cdot f_{22}''$$

如果 f 的二阶偏导数连续，则 $f_{12}'' = f_{21}''$

四、全微分的形式不变性

以 $z = f(u,v)$，$u = \varphi(x,y)$，$v = \psi(x,y)$ $\Rightarrow$ $z = f[\varphi(x,y), \psi(x,y)] \triangleq z(x,y)$ 为例，

则

$$dz = \frac{\partial z}{\partial x} \cdot dx + \frac{\partial z}{\partial y} \cdot dy$$
$$= \left(\frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x} \right) \cdot dx + \left(\frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial y} \right) \cdot dy$$
$$= \frac{\partial f}{\partial u} \cdot \left(\frac{\partial u}{\partial x} \cdot dx + \frac{\partial u}{\partial y} \cdot dy \right) + \frac{\partial f}{\partial v} \cdot \left(\frac{\partial v}{\partial x} \cdot dx + \frac{\partial v}{\partial y} \cdot dy \right)$$
$$= \frac{\partial f}{\partial u} \cdot du + \frac{\partial f}{\partial v} \cdot dv$$